The Shortest Route Can Have More Steps

Explain it in your own words

Use this guide to understand each animation, choose your phrasing, and keep a natural YouTube delivery. Every clip has a matching frame, the idea behind the movement, and a specific delivery cue.

16 clips | 09:33.6 continuous video | 10 guide pages

A simple speaking rhythm

1. Set up: Tell viewers what the objects represent and ask one small question.

2. Follow the change: Point to what moves, grows, disappears or gets replaced.

3. Explain the cause: Say why that change happened. Use the mechanism notes as your checklist.

4. Let it land: Pause on the result, then connect it to the next question.

Use the timeline flexibly

The next page gives exact clip boundaries in the current continuous MP4. Individual clip files begin at 00:00. In the self-paced player, a phase holds until you advance, so your own recording can be longer or shorter. These timestamps describe the video; delivery cues describe how you might explain it.

Press Right / Left to move between phases, Up / Down to adjust speed, R to replay, and Shift+P for the separate script window. Rehearse the cause-and-effect in your own words before recording.

Open this video in the local player | Full narration reference | Sources

The complete timeline

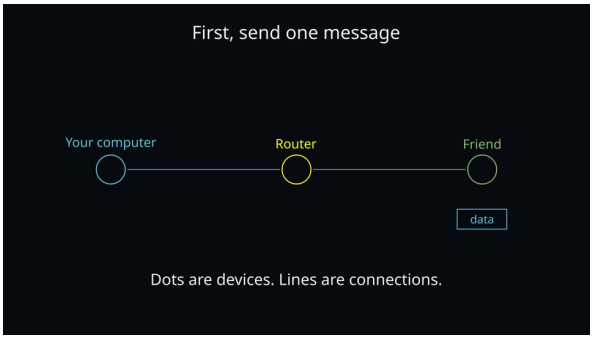
Find the clip, see the question being explained, and jump to its guide page. Times are MM:SS.t in the full MP4.

CLIP / TIME	WHAT YOU ARE EXPLAINING	PAGE
01 00:00.0 - 00:35.2	First, send one message	3
02 00:35.2 - 01:10.4	Give each step a price	3
03 01:10.4 - 01:44.3	Grow a small map	4
04 01:44.3 - 02:19.9	Keep a little scorecard	4
05 02:19.9 - 02:54.7	Look one step from home	5
06 02:54.7 - 03:30.4	A better way to B	5
07 03:30.4 - 04:06.9	Choose the cheapest unfinished dot	6
08 04:06.9 - 04:42.1	Find a first route to F	6
09 04:42.1 - 05:18.2	One last improvement	7
10 05:18.2 - 05:54.7	Know when the answer is ready	7
11 05:54.7 - 06:31.3	Why the rule works	8
12 06:31.3 - 07:06.9	Remember the way back	8
13 07:06.9 - 07:43.0	Walk the winning route	9
14 07:43.0 - 08:19.1	What if a connection breaks?	9
15 08:19.1 - 08:56.6	Find the new winner	10
16 08:56.6 - 09:33.6	A big answer from small steps	10

Use the clip number as your anchor when recording. Matching frames show the completed diagram; watch the movement too, because the change is often the explanation.

CLIP 01 / First, send one message

00:00.0 - 00:35.2 in full video | 35.2s clip



What the viewer sees
A message passes through devices drawn as dots and lines.

What you need to explain - and why
Routers pass messages onward through connections. A route is a sequence of those connections from a start to a destination.

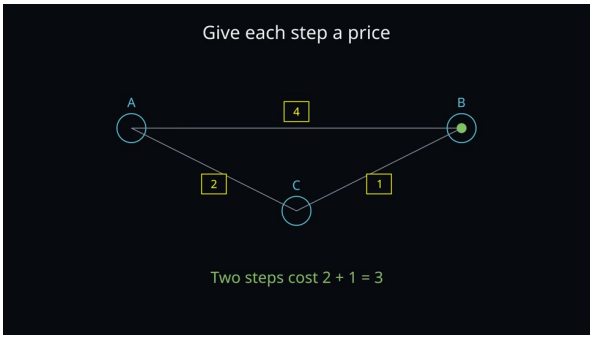
Tone and emphasis
Build this tiny map before mentioning algorithms.

Next: Give each step a price

Open clip 01 | My phrasing: _____

CLIP 02 / Give each step a price

00:35.2 - 01:10.4 in full video | 35.2s clip



What the viewer sees
A direct connection costs four while a two-step route costs three.

What you need to explain - and why
Route cost is the sum of connection costs. Fewest steps and lowest total can give different winners.

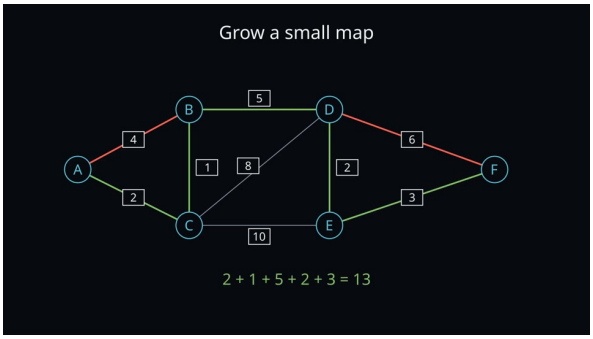
Tone and emphasis
Ask which route is cheaper before adding two and one.

Next: Grow a small map

Open clip 02 | My phrasing: _____

CLIP 03 / Grow a small map

01:10.4 - 01:44.3 in full video | 33.9s clip



What the viewer sees
The map grows and two routes to F are compared.

What you need to explain - and why
The three-step route totals 15, while a five-step route totals 13. A repeatable method helps when comparing routes by eye becomes difficult.

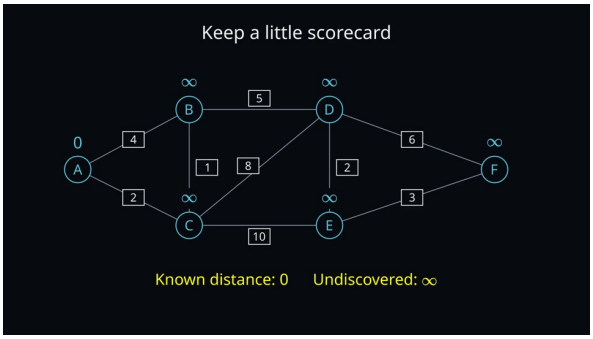
Tone and emphasis
Introduce Dijkstra as the recipe for this puzzle.

Next: Keep a little scorecard

Open clip 03 | My phrasing: _____

CLIP 04 / Keep a little scorecard

01:44.3 - 02:19.9 in full video | 35.7s clip



What the viewer sees
Scores begin at zero for A and unknown for other dots.

What you need to explain - and why
A score records the cheapest route found so far. The infinity symbol is a bookkeeping marker for no route discovered yet.

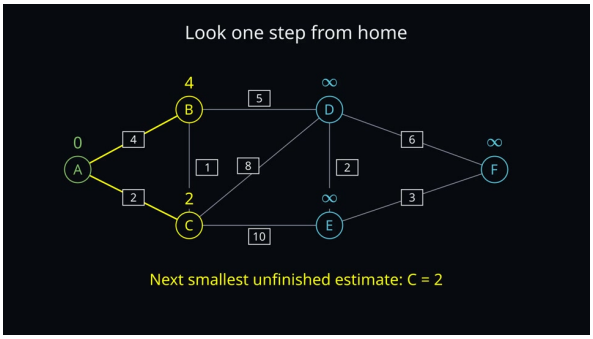
Tone and emphasis
Explain the question marks before introducing infinity.

Next: Look one step from home

Open clip 04 | My phrasing: _____

CLIP 05 / Look one step from home

02:19.9 - 02:54.7 in full video | 34.8s clip



What the viewer sees
A's neighbors receive scores four and two.

What you need to explain - and why
Starting at A, add each available connection cost. Choose the smallest unfinished score next: C at two.

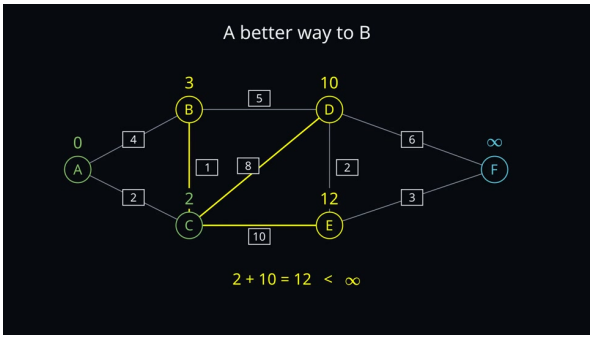
Tone and emphasis
Say which score is known, which is unfinished, and what green will mean.

Next: A better way to B

Open clip 05 | My phrasing: _____

CLIP 06 / A better way to B

02:54.7 - 03:30.4 in full video | 35.7s clip



What the viewer sees
Routes through C improve B and discover D and E.

What you need to explain - and why
Reaching C already cost two. Adding C-B's one gives three, which improves B's four; other additions give D ten and E twelve.

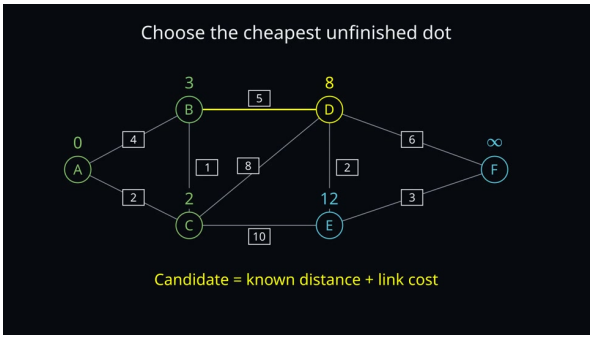
Tone and emphasis
Work one comparison fully before moving through the remaining neighbors.

Next: Choose the cheapest unfinished dot

Open clip 06 | My phrasing: _____

CLIP 07 / Choose the cheapest unfinished dot

03:30.4 - 04:06.9 in full video | 36.5s clip



What the viewer sees
B's smaller unfinished score improves D.

What you need to explain - and why
A route via B costs three so far plus five for the next connection, giving eight. Eight replaces ten.

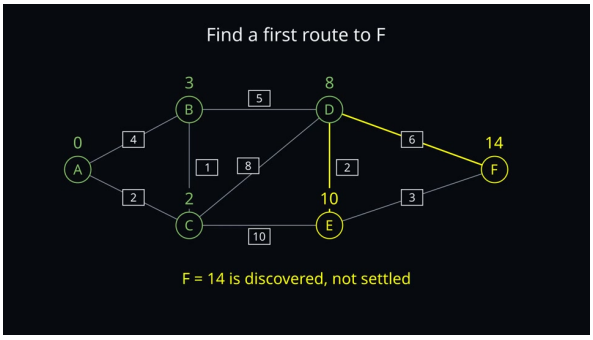
Tone and emphasis
Stress adding to the whole journey, not choosing the cheapest single edge.

Next: Find a first route to F

Open clip 07 | My phrasing: _____

CLIP 08 / Find a first route to F

04:06.9 - 04:42.1 in full video | 35.2s clip



What the viewer sees
D improves E and discovers a route to F.

What you need to explain - and why
Eight plus two improves E to ten; eight plus six gives F fourteen. Discovering F does not yet make fourteen final.

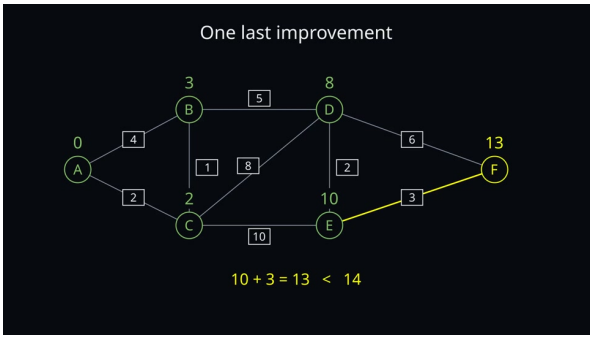
Tone and emphasis
Pause on the apparent finish, then ask whether a cheaper route may remain.

Next: One last improvement

Open clip 08 | My phrasing: _____

CLIP 09 / One last improvement

04:42.1 - 05:18.2 in full video | 36.1s clip



What the viewer sees
E improves F from fourteen to thirteen.

What you need to explain - and why
Ten to reach E plus three more beats the previous fourteen. The same update rule produces the better answer.

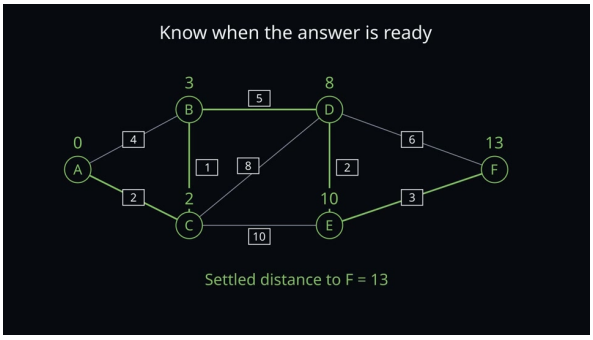
Tone and emphasis
Let this be the payoff for refusing to stop too early.

Next: Know when the answer is ready

Open clip 09 | My phrasing: _____

CLIP 10 / Know when the answer is ready

05:18.2 - 05:54.7 in full video | 36.5s clip



What the viewer sees
F becomes the smallest unfinished score and is finalized.

What you need to explain - and why
With nonnegative costs, selecting the smallest unfinished score makes it final. For one destination, we can stop when that destination is selected.

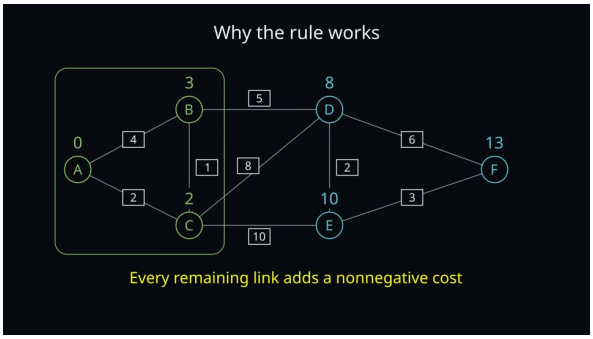
Tone and emphasis
Separate found an answer from know the answer is final.

Next: Why the rule works

Open clip 10 | My phrasing: _____

CLIP 11 / Why the rule works

05:54.7 - 06:31.3 in full video | 36.5s clip



What the viewer sees
A frontier highlights why adding more cost cannot beat a smaller score.

What you need to explain - and why
Any path reaching an unfinished point already costing ten cannot beat eight by adding nonnegative costs. Negative costs would break this argument.

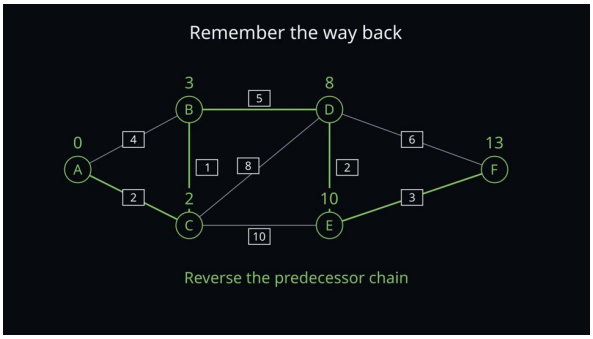
Tone and emphasis
Give the small eight-versus-ten comparison before stating the restriction.

Next: Remember the way back

Open clip 11 | My phrasing: _____

CLIP 12 / Remember the way back

06:31.3 - 07:06.9 in full video | 35.7s clip



What the viewer sees
A chain is followed backward from F to A.

What you need to explain - and why
Remember the predecessor whenever a score improves. Following those reminders backward reconstructs the route, then reversing gives travel order.

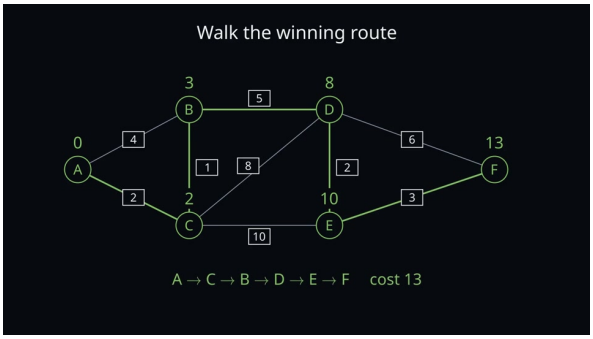
Tone and emphasis
Describe reminders first, then use predecessor as an optional name.

Next: Walk the winning route

Open clip 12 | My phrasing: _____

CLIP 13 / Walk the winning route

07:06.9 - 07:43.0 in full video | 36.1s clip



What the viewer sees
The winning route is traversed with a running total.

What you need to explain - and why
A-C-B-D-E-F costs 2 + 1 + 5 + 2 + 3 = 13. Shape and number of lines do not determine the winner.

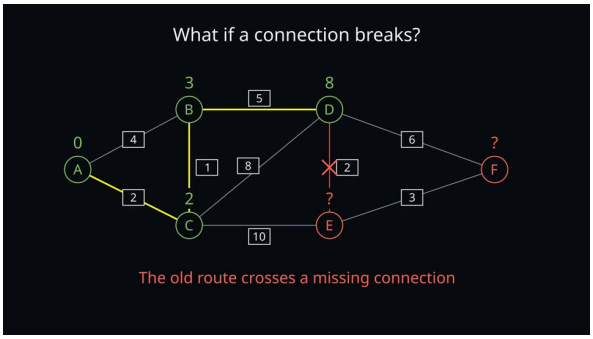
Tone and emphasis
Count each step aloud with the moving marker.

Next: What if a connection breaks?

Open clip 13 | My phrasing: _____

CLIP 14 / What if a connection breaks?

07:43.0 - 08:19.1 in full video | 36.1s clip



What the viewer sees
The D-E connection is removed.

What you need to explain - and why
The old winning route is no longer valid. Changing the available connections requires a new calculation on the updated map.

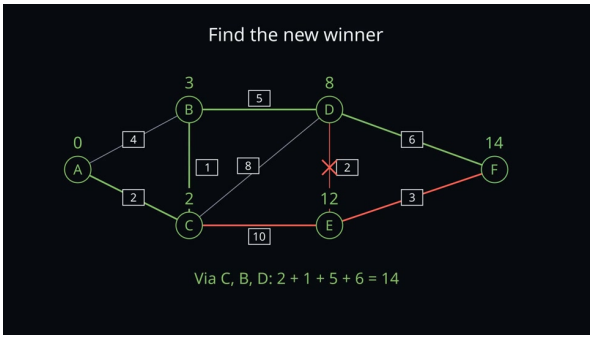
Tone and emphasis
Invite a new prediction rather than pretending the old answer still works.

Next: Find the new winner

Open clip 14 | My phrasing: _____

CLIP 15 / Find the new winner

08:19.1 - 08:56.6 in full video | 37.4s clip



What the viewer sees
Two replacement routes compete after the failure.

What you need to explain - and why
A-C-B-D-F now totals 14; A-C-E-F totals 15. The same recipe finds new final scores, including E twelve and F fourteen.

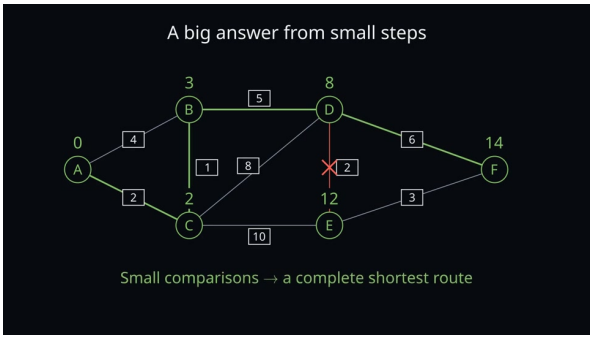
Tone and emphasis
Compare totals with the old connection visibly absent.

Next: A big answer from small steps

Open clip 15 | My phrasing: _____

CLIP 16 / A big answer from small steps

08:56.6 - 09:33.6 in full video | 37.0s clip



What the viewer sees
The updated route completes the story.

What you need to explain - and why
Small score improvements and choosing the cheapest unfinished point build a complete least-cost route.

Tone and emphasis
Recap the routine in your own verbs: choose, add, compare, remember.

Close: return to the opening question and its answer.

Open clip 16 | My phrasing: _____